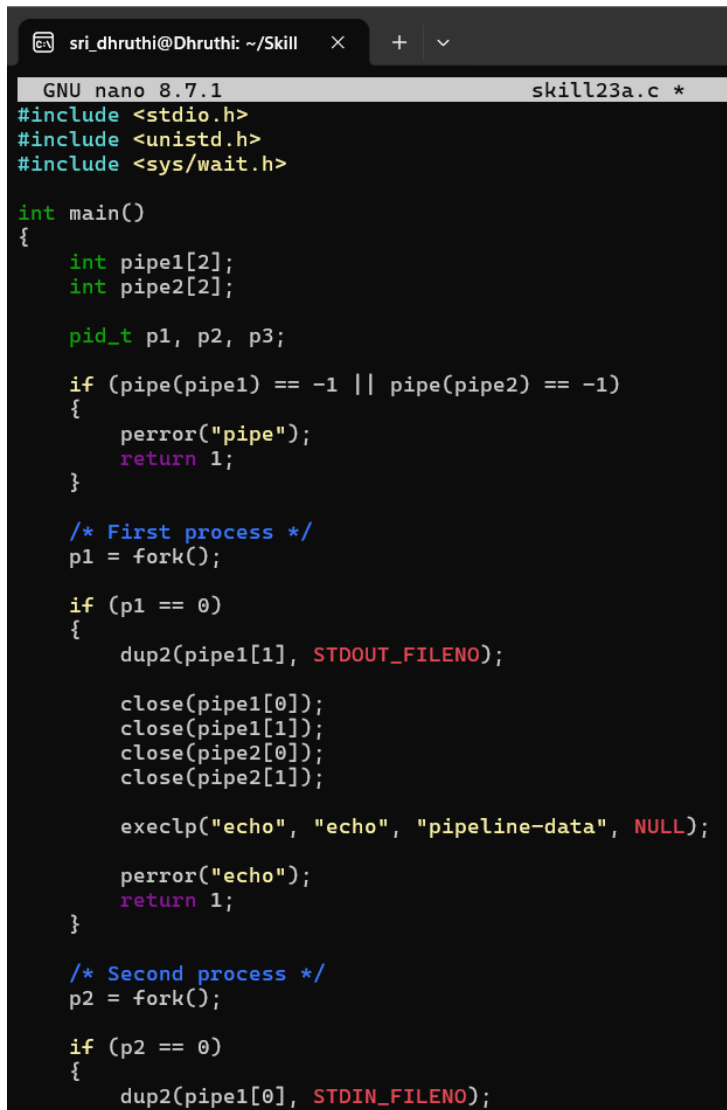


Os skill 23

23(a) Execute Large Pipelines

What it does: Creates a three-stage pipeline using multiple processes and pipes, then waits for all processes to finish.



```
GNU nano 8.7.1 skill23a.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipe1[2];
    int pipe2[2];

    pid_t p1, p2, p3;

    if (pipe(pipe1) == -1 || pipe(pipe2) == -1)
    {
        perror("pipe");
        return 1;
    }

    /* First process */
    p1 = fork();

    if (p1 == 0)
    {
        dup2(pipe1[1], STDOUT_FILENO);

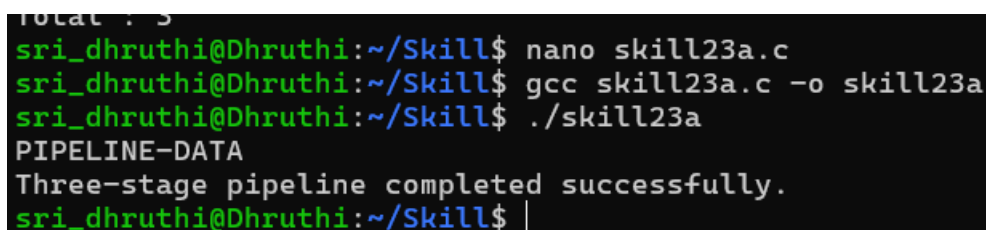
        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("echo", "echo", "pipeline-data", NULL);

        perror("echo");
        return 1;
    }

    /* Second process */
    p2 = fork();

    if (p2 == 0)
    {
        dup2(pipe1[0], STDIN_FILENO);
```



```
Total : 3
sri_dhruthi@Dhruthi:~/Skill$ nano skill23a.c
sri_dhruthi@Dhruthi:~/Skill$ gcc skill23a.c -o skill23a
sri_dhruthi@Dhruthi:~/Skill$ ./skill23a
PIPELINE-DATA
Three-stage pipeline completed successfully.
sri_dhruthi@Dhruthi:~/Skill$ |
```

23(b) Document Execution Path and System Calls

What it does: Demonstrates the basic shell execution path using fork(), exec(), and waitpid().

```
sri_dhruthi@Dhruthi: ~/Skill × + v
GNU nano 8.7.1 skill23b.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child: executing command.\n");

        execlp("echo",
               "echo",
               "Syscall path: fork -> exec -> exit",
               NULL);

        perror("exec");
        return 1;
    }

    printf("Parent: fork created child PID = %d\n", pid);

    waitpid(pid, NULL, 0);

    printf("Parent: wait completed.\n");
    printf("Execution path finished successfully.\n");

    return 0;
}
```

```
Three-stage pipeline completed successfully.
sri_dhruthi@Dhruthi:~/Skill$ nano skill23b.c
sri_dhruthi@Dhruthi:~/Skill$ gcc skill23b.c -o skill23b
sri_dhruthi@Dhruthi:~/Skill$ ./skill23b
Parent: fork created child PID = 1063
Child: executing command.
Syscall path: fork -> exec -> exit
Parent: wait completed.
Execution path finished successfully.
sri_dhruthi@Dhruthi:~/Skill$ |
```

